FABER-CASTELL
Date
Page No.

Lecture-11

- 2 popula^n with means μ_1 & μ_2, variances σ_1^2 & σ_2^2. We sample $X_1, \dots, X_n$ & $Y_1, \dots, Y_n$. The 2 popula^{ns} must be independent.
  We want to estimate $\mu_1 - \mu_2$.
  $\bar{X} - \bar{Y}$

- $E[\bar{X}] = \mu_1$, $E[\bar{Y}] = \mu_2$
  $\therefore$, $E[\bar{X} - \bar{Y}] = E[\bar{X}] - E[\bar{Y}] = \mu_1 - \mu_2$
  $\therefore$, $(\mu_1 - \mu_2)$ has unbiased estimator $(\bar{X} - \bar{Y})$.

- $Var(\bar{X}) = \dfrac{\sigma_1^2}{n_1}$, $Var(\bar{Y}) = \dfrac{\sigma_2^2}{n_2}$
  $Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) - 2\,Cov(\bar{X}, \bar{Y})$
  since they are independent cov $= 0$

$$\boxed{Var(\bar{X} - \bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$\boxed{SE(\bar{X} - \bar{Y}) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- if popula^n var. is known
  $\bar{X} \sim N\left(\mu_1, \dfrac{\sigma_1^2}{n_1}\right)$, $\bar{Y} \sim N\left(\mu_2, \dfrac{\sigma_2^2}{n_2}\right)$
  $(\bar{X} - \bar{Y}) \sim N\left(\mu_1 - \mu_2, \dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}\right)$

  for $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$

$$\boxed{(\bar{X} - \bar{Y}) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$ CI for $\mu_1 - \mu_2$

Eg. $n_A = 50$, $\bar{X}_A = 36$, $\sigma_A = 6$ 95% CI $\alpha = 0.05$
$n_B = 75$, $\bar{X}_B = 42$, $\sigma_B = 8$ $\mu_B - \mu_A$ $\alpha/2 = 0.025$
$z_{\alpha/2} = 1.96$

$$(42 - 36) \pm 1.96\underbrace{\sqrt{\frac{36}{50} + \frac{64}{75}}}$$

$\sqrt{0.72 + 0.85}$
$\Rightarrow 1.256$

FABER-CASTELL
Date
Page No.

$(42-36) \pm (1.96)(1.257) \rightarrow (42-36) \pm 2.46372$

$6 \pm 2.46372 \Rightarrow (3.53628, 8.46372)$

- what if popula^n var. is unknown $\rightarrow \sigma_1^2, \sigma_2^2$

$\sigma_1^2 = \sigma_2^2$ — equal, but unknown

$\sigma_1^2 \neq \sigma_2^2$ — unequal & unknown

- equal var but unknown

$\sigma_1^2 = \sigma_2^2 = \sigma^2$ but $s_1^2 \neq s_2^2$

So, we can pool the var.

$$\boxed{s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

$$\frac{(n_1-1)s_1^2}{\sigma^2} \sim \chi^2_{n_1-1} \quad \& \quad \frac{(n_2-1)s_2^2}{\sigma^2} \sim \chi^2_{n_2-1}$$

since 2 are independent,

$$\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}$$

So, if common σ is known,

$$\frac{(\bar{X}_1-\bar{X}_2) - (\mu_1-\mu_2)}{\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim N(0,1)$$

σ is unknown

$$T = \frac{(\bar{X}_1-\bar{X}_2) - (\mu_1-\mu_2)}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

$\therefore$, CI is $\boxed{(\bar{X}_1-\bar{X}_2) \pm t_{\alpha/2,\, n_1+n_2-2}\, s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$

eg. $n_1 = 12$, $\bar{x}_1 = 3.11$, $s_1 = 0.771$ — 90% $\alpha = 0.1$ $\alpha/2 = 0.05$

$n_2 = 10$, $\bar{x}_2 = 2.04$, $s_2 = 0.448$ — $(\mu_1-\mu_2)$

popula^n has equal var.

I mistake s_1^2

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{20} = \frac{11(0.771) + 9(0.448)}{21}$$

$$= \frac{8.481 + 4.032}{20} = 0.62565$$

$s_p = 0.791$

$$\bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2,\, n_1+n_2-2}\; S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\underbrace{(3.11 - 2.04)}_{1.07} \pm \underbrace{t_{0.05,\,20}}_{2.086} (0.79)(0.424)$$

$$1.07 \pm 0.6987 \Rightarrow (0.3713,\ 1.7687)$$

- σ_1^2, σ_2^2 both unknown & $\sigma_1^2 \neq \sigma_2^2$
  we can't pool sample var.

$$\boxed{SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \qquad T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}}$$

we can't use $n_1 + n_2 - 2$, we have to use satterthwaite approx.

welch 2-sample t procedure

$$\boxed{\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left(\frac{s_1^2/n_1}{n_1-1}\right)^2 + \left(\frac{s_2^2/n_2}{n_2-1}\right)^2}}$$

interval is,

$$\boxed{(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2,\,\nu} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Table

	SE	dist
σ_1^2, σ_2^2 known	$\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$	z
$\sigma_1^2 = \sigma_2^2$, unknown	$S_p\sqrt{1/n_1 + 1/n_2}$	$t_{\alpha/2,\, n_1+n_2-2}$
$\sigma_1^2 \neq \sigma_2^2$, unknown	$\sqrt{s_1^2/n_1 + s_2^2/n_2}$	t_ν

Point est. $\pm$ critical value $\times$ SE (point estimate)

FABER-CASTELL
Date
Page No.

- X_i, Y_i within pair they are dependent. Across pairs, they are independent.

$\boxed{D_i = X_i - Y_i}$ $\Rightarrow D_1, D_2 \ldots D_n$

$\mu_D = E[D]$ if $D = X - Y$ then $\mu_D = \mu_X - \mu_Y$

$\bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i$. $S_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(D_i - \bar{D})^2$

then, $\frac{\bar{D} - \mu_D}{S_D/\sqrt{n}} \sim t_{n-1}$

$\therefore$, $100(1-\alpha)\%$ for μ_D is $\boxed{\bar{D} \pm t_{\alpha/2, n-1}\frac{S_D}{\sqrt{n}}}$

Eg. 20 soldiers (p_i, f_i) $D_i = (p_i - f_i)$

$\bar{d} = -0.87$, $S_d = 2.9773$, $df = 19$
$\alpha = 0.05$, $\alpha/2 = 0.025$

$$-0.87 \pm \underbrace{t_{0.025, 19}}_{2.093} \left.\frac{(2.9773)}{\sqrt{19} = 4.358}\right\} 0.683$$

$-0.87 \pm (2.093)(0.683) \Rightarrow -0.87 \pm 1.43 \Rightarrow (-2.3, 0.56)$

Eg. $X_i \Rightarrow$ no of rooms, Y_i = area $X_i - Y_i$
$(X_1, Y_1) \Rightarrow$ ravi & yash marks for same subj. $X_i - Y_i$

Date
Page No.

- DT

if paired $\quad D_i = X_i - Y_i$

$$\bar{D} \pm t_{\alpha/2, n-1} \frac{S_D}{\sqrt{n}}$$

if independent

is population var. known

Yes $\Rightarrow$ $(\bar{X}_1 - \bar{X}_2) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

no

equal var.

yes $\Rightarrow$ $S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1 + n_2 - 2} \, S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

no $\Rightarrow$ $\quad \nu \Rightarrow$ by welch

$$\bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2, \nu} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$